

SPSS 7 Factor Analysis

01/07/2015

1

Chapter 7 Factor Analysis

- Agenda
 - Why using factor analysis
 - Principle of factor analysis
 - Steps of factor analysis
 - Reliability analysis
 - Assignments

01/07/2015

2

Why using factor analysis?

- For most research exercises, many variables are measured. However, some of them are correlated, some are repeated measurement, some irrelevant. It is difficult to interpret the result and determine the significant factors, without extraction and grouping of variables.
- eg. Raymond B. Catell initially had 4500 measures on character traits. After factor analysis, he reduced the measures to 200 questions that can quantify 16 different personality traits. It is called 16 Personality Factor Questionnaire (16PF).

01/07/2015

3

Usage of Factor analysis

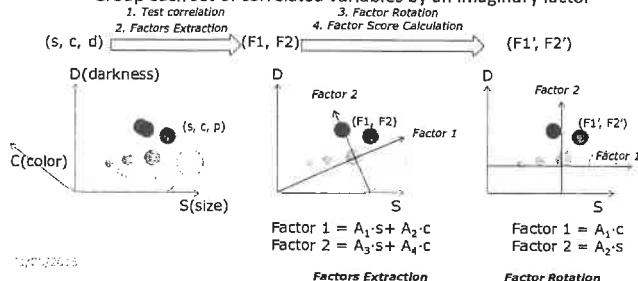
- Linear combination
 - Regression line, factor model, factor structure matrix
- Grouping of variables into factors, reduce dimensions
 - How many factors? Rotation, factor correlation matrix
- Simplify cases
 - Compute factor scores for each case
- Model the observed data
 - Residue, reproduced correlation
- Explain the variance
 - Eigenvalues, scree plot, communality, principal component analysis (PCA)

01/07/2015

4

Factor analysis

- Construct a model of the data that:
 - Reveal a simple and ideal pattern using as few factor as possible
 - Explain the variance as much as possible
 - Group each set of correlated variables by an imaginary factor



01/07/2015

5

Example of This Session

- Data File: gettingahead.sav
- Data:
 - Questionnaire on respondent's view on the requirements to achieve higher social status
 - Variables, each represent a question:

wealth – need wealthy family	clout – need political connection
appearance – need educated parents	race – need to be right race
edself – need good education	religion – need to be right religion
ambition – need ambition	region – must be from right region
ability – need natural ability	gender – need to be right sex
hard work – need to work hard	pol belief – must have certain political view
knowright – need to know right people	

01/07/2015

6

Step1 Compute Correlation Matrix

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy		
		.820
Bartlett's Test of Sphericity		
Approx. Chi-Square		4085.321
df		78
Sig.		.000

- KMO measure
 - KMO = .820 means most variables are linearly related and can go ahead with the factor analysis
- Bartlett's test of sphericity
 - Test the null hypothesis that the observed data are from a multivariate normal population in which all correlation coefficients are 0.

6/10/2015

10

Step1 Compute Correlation Matrix

- Anti-image matrices
 - Covariance matrix
 - Correlation matrix
- Get the sampling adequacy from matrix diagonal entries
 - The values are reasonably large
 - Remove variables with small values, if any

Initial Eigenvalues														
Component	Total	Variance	% of Variance	Cumulative %	Total	Variance	% of Variance	Cumulative %	Total	Variance	% of Variance	Cumulative %	Total	Variance
1	3.822	29.402	29.402	29.402	1.776	12.660	12.660	12.660	1.213	9.332	9.332	9.332	.553	7.739
2	1.776	12.660	12.660	42.062	.793	6.099	6.099	18.762	.752	5.788	5.788	24.550	.684	5.104
3	1.213	9.332	9.332	51.394	.684	5.104	5.104	29.866	.654	5.029	4.979	34.845	.592	4.479
4	.553	7.739	7.739	59.133	.592	4.479	4.479	39.324	.496	3.814	3.814	43.138	.480	3.755
5	.793	6.099	6.099	65.232	.442	3.401	3.401	46.539	.442	3.401	3.401	49.940	.385	2.809
6	.752	5.788	5.788	71.020										
7	.684	5.104	5.104	76.124										
8	.654	5.029	4.979	81.143										
9	.592	4.479		85.622										
10	.496	3.814		90.436										
11	.480	3.755		94.191										
12	.442	3.401		97.592										
13	.385	2.809		100.000										

Variable Name	Sampling Adequacy
wealth	.850
edparan	.780
edself	.844
ambition	.837
ability	.835
hardwork	.573
knowright	.821
clout	.835
race	.874
religion	.810
region	.869
gender	.859
jobbelief	.840

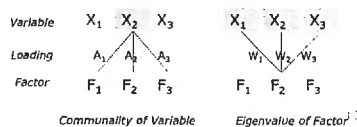
6/10/2015

11

Step 2 Extract The Factors

- Factor analysis assumptions
 - Underlining factors are responsible for the variable correlation
 - Correlated variables share common factors
- Sample size
 - Large enough to extract "real" factors, 10 ~ 300 cases
- Factor model
 - Like the multiple regression equation
 - $X = A_1F_1 + A_2F_2 + \dots + u$ (each variable is a linear combination of all factors)
 - F_i – common factors
 - A_i – coefficient / loadings
 - u – unique factor, specific to variable, and include error
 - $F = W_1X_1 + W_2X_2 + \dots + W_nX_n$ (each factor is linear combination of all variables)
 - F – factor
 - W_i – coefficient / weightings
 - X_i – variables

$$\begin{aligned} [A] * [F] &= [X] \\ [F] &= [A^{-1} * X] \\ [A^{-1} * W] & \end{aligned}$$



6/10/2015

12

Step 2 Extract The Factors

- Method of factor extraction
 - Principle components analysis (PCA)
 - uses an **orthogonal transformation** to convert a set of observations of **correlated variables** into a set of values of **uncorrelated variables** called **principal components**.
 - Reduce variables
 - Reveal the internal structure of the data in the way that best explain the variance of data
 - "Component" and "factor" are different words with the same meaning

Step 2 Extract The Factors

- Principle components analysis (PCA) (Con't)
 - First principle component
 - the combination that explain the largest amount of variance
 - Second principle component
 - the combination that explain the second largest amount of variance,
 - uncorrelated to the first principle component
 - Third principle component...

6/10/2015

13

Step 2 Extract The Factors

- Eigenvalue
 - The proportion of variance explained by each factor
- Total variance explained
 - % of variance column shows Factor 1 explains 29% of total variance
 - Cumulative % column shows first 3 factors explains over 50% of total variance
 - Total column shows first 3 factors explains more than one variance

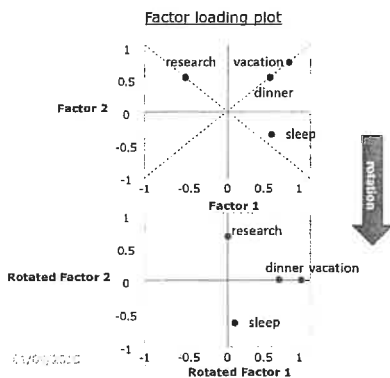
Component	Total Variance Explained		
	Total	% of Variance	Cumulative %
1	3.822	29.402	29.402
2	1.776	12.660	42.062
3	1.213	9.332	51.394
4	.553	7.739	59.133
5	.793	6.099	65.232
6	.752	5.788	71.020
7	.684	5.104	76.124
8	.654	5.029	81.143
9	.592	4.479	85.622
10	.496	3.814	90.436
11	.480	3.755	94.191
12	.442	3.401	97.592
13	.385	2.809	100.000

Extraction Method: Principal Component Analysis.

6/10/2015

14

- Factor loading plot: Coordinate of each variable are the correlation coefficients (factor loading) with 2 factors. E.g. Sleep is at (0.5, -0.4)
- Before rotation: all variables are related to all factors, difficult to interpret the factors
- After rotation: simpler structure between variables and factors, easy to interpret the factors
 - Eg. - Factor 1: Dinner, vacation → hedonism
 - Factor 2: sleep, research → scholarship



Step 3 Rotate The Factors

- Rotated Component Matrix
 - Use **Varimax** Orthogonal Rotation, which minimizes the number of variables loading on a factor
 - With **Options** → **sorted by sizes selected**, the highest factor loadings for each factor are selected and listed in separate blocks
 - The factor loading within each block are sorted
 - Useful to depict the group of variables influenced by each factor

Rotated Component Matrix^a

	Component		
	1	2	3
region	.768	.193	.019
religion	.764	.086	.017
polbelief	.743	.188	.024
gender	.641	.216	-.034
race	.631	.316	-.055
wealth	.193	.734	-.055
edparen	.040	.794	.134
knowright	.389	.658	.122
clout	.432	.601	.018
hardwork	.040	-.184	.781
ambition	-.078	.020	.715
edself	-.132	.356	.588
ability	.206	.252	.404

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.
a. Rotation converged in 5 iterations.

Step 3 Rotate The Factors

- Rotated Component Matrix
 - For easy interpretation of the factor, use **Options** → **Suppress small coefficients** and set the value below which the factors are not shown

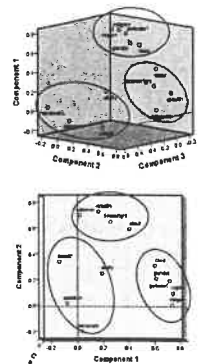
Rotated Component Matrix^a

	Component		
	1	2	3
region	.768		
religion	.764		
polbelief	.743		
gender	.641		
race	.631		
wealth		.734	
edparen		.794	
knowright		.658	
clout		.601	
hardwork			.781
ambition			.715
edself			.588
ability			.404

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.
a. Rotation converged in 5 iterations.

Step 3 Rotate The Factors

- Component plot in rotated space
 - Good for examine the success of orthogonal rotation
 - For more than 3 factors, SPSS only provide 3-dimensional plot
 - To change zoom in the plot:
 - Edit > select X axis > **scale>range**
 - set minimum to **-0.2**
 - set maximum to **0.8**
 - set major increment to **0.2**
 - >apply > close
 - Repeat the above steps to change Y and Z axis
 - To 3D rotate the plot
 - Edit > 3D rotation
 - Use mouse to drag to rotate the view
 - Rotate to a 2-dimension view is good for checking the relation of the 2 factors
 - Ideally clusters of variables should be at the tips of axes



Step 4: Compute Factor Scores

- Factor scores
 - Factor scores for each case is the weighted sum of all variables of a case
 - $F_1 = W_1(\text{ability}) + W_2(\text{ambition}) + \dots + W_{13}(\text{wealth})$
 - Shows the overall performance of each case
 - 3 methods: regression, Anderson-Rubin, Bartlett
 - After factor analysis, factor scores are automatically created in **Data Editor**
- Use factor scores to test hypothesis
 - eg. t-test between groups for a factor
 - Require further reading on the topic

Component Score Coefficient Matrix

	Component		
	1	2	3
ability	.046	.058	.293
ambition	.011	-.041	.442
clout	.049	.246	-.035
edparen	-.143	.386	.088
edself	-.116	.176	.311
gender	.229	-.021	-.018
hardwork	.064	-.202	.512
knowright	-.027	.204	.018
polbelief	.281	-.053	.025
race	.261	.041	-.042
region	.314	-.123	.034
religion	.325	-.170	.013
wealth	-.091	.388	-.107

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.
Component Scores.

Step 5: Interpretation of Results

- Interpretation
 - From the rotated component matrix, it is obvious that there are 3 groups of variables, each of which are represented by a factor
 - Factor 1 contains region, religion, polbelief, gender, race
 - Factor 2 contains wealth, edparen, knowright, clout
 - Factor 3 is hardwork, ambition, edself, ability
- Conclusion
 - By looking at the themes of the factors, Factor 1 is identified as social attributes, Factor 2 represents family factor, and Factor 3 is personal achievement

Rotated Component Matrix^a

	Component		
	1	2	3
region	.768	.193	.019
religion	.764	.086	.017
polbelief	.743	.188	.024
gender	.641	.216	-.034
race	.631	.316	-.055
wealth	.193	.734	-.055
edparen	.040	.794	.134
knowright	.389	.658	.122
clout	.432	.601	.018
hardwork	.040	-.184	.781
ambition	-.078	.020	.715
edself	-.132	.356	.588
ability	.206	.252	.404

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.
a. Rotation converged in 5 iterations.

Results on Reliability Analysis

- Reliability Statistics
 - Cronbach's α (0.778)
 - > 0.7 for good reliability
 - Set to > 0.8 for cognitive tests
- Inter-Item Correlation Matrix

Reliability Statistics		
Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.778	.761	13

- The same as the correlation matrix in factor analysis
- Depict the coefficients of correlation among variables

6/15/2015

47

Results on Reliability Analysis

- Item-Total Statistics
 - "Corrected Item-Total Correlation" shows the correlation between the item and the total score
 - Look for values < 0.3 which has low correlation and may have to be dropped

Item-Total Statistics					
Item	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item Total Correlation	Squared Multiple Correlation	Corrected Total
wealth	31.57	22.764	.484	.324	718
education	35.95	25.464	.390	.237	745
income	37.88	26.637	.450	.320	785
employment	37.88	26.637	.276	.164	785
family	36.47	27.441	.391	.378	773
employment	37.88	26.637	.648	.576	790
employment	36.13	26.819	.574	.387	754
education	33.34	23.123	.348	.228	747
income	35.11	23.271	.331	.212	758
employment	36.86	23.918	.414	.282	788
region	34.14	24.863	.573	.380	752
gender	35.10	23.549	.477	.337	755
parental	35.80	23.425	.581	.423	768

6/15/2015

48

Results on Reliability Analysis

- Item-Total Statistics (Con't)
 - "Cronbach's Alpha if Item Deleted"
 - α after deletion of the item
 - If the value is higher than the overall α (0.778), it may be worth to delete the item to improve reliability
 - Rerun factor analysis after deletion to check the factor structure

refer

Item-Total Statistics					
Item	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item Total Correlation	Squared Multiple Correlation	Corrected Total
wealth	31.57	22.764	.484	.324	718
education	35.95	25.464	.390	.237	745
income	37.88	26.637	.450	.320	785
employment	37.88	26.637	.276	.164	785
family	36.47	27.441	.391	.378	773
employment	37.88	26.637	.648	.576	790
employment	36.13	26.819	.574	.387	754
education	33.34	23.123	.348	.228	747
income	35.11	23.271	.331	.212	758
employment	36.86	23.918	.414	.282	788
region	34.14	24.863	.573	.380	752
gender	35.10	23.549	.477	.337	755
parental	35.80	23.425	.581	.423	768

6/15/2015

49

Task: Split-Half Reliability

- To perform split-half reliability analysis:
 - Recall the Reliability Analysis dialogue
 - Set Model to Split-half
 - > OK

6/15/2015

50

Result of Split-Half Reliability

- The items are separated into 2 equal parts for calculation of α
 - The grouping depends on the sequence of selection of the items
- For reliable items, the two α should be close.
- Note the grouping may affect the results.

Reliability Statistics		
Cronbach's Alpha	Part 1	Value
	Part 2	Value
Cronbach's Alpha	Part 1	Value
	Part 2	Value
Total # of Items		
Spearman-Brown Formula		
Coefficient		
Guttman Split-Half Coefficient		

a. The items are: wealth, education, income, employment, family, employment, employment, education, income, employment, region, gender, parental.

6/15/2015

51

Assignment 2

- Continue with "helping2.sav"
- Requirements:
 - Analyze the reliability of the survey using Cronbach's α and Split-half analysis
 - Report and discuss the process and result of the analysis

6/15/2015

52